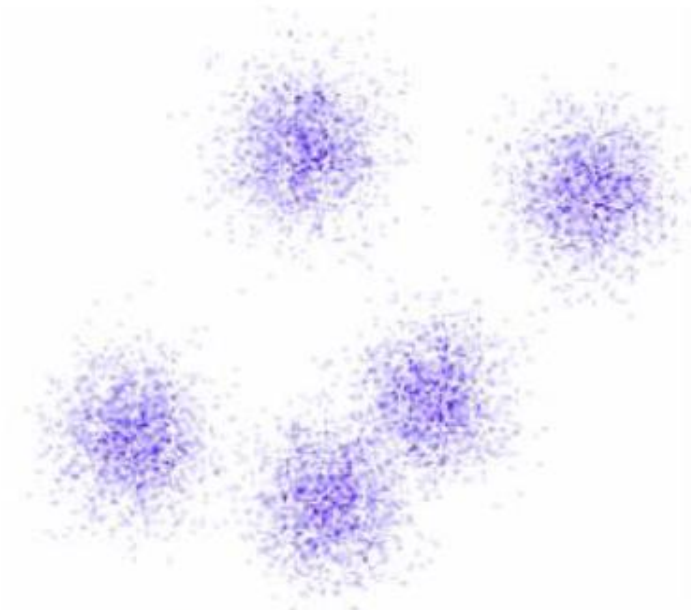
Spectral Clustering Introduction

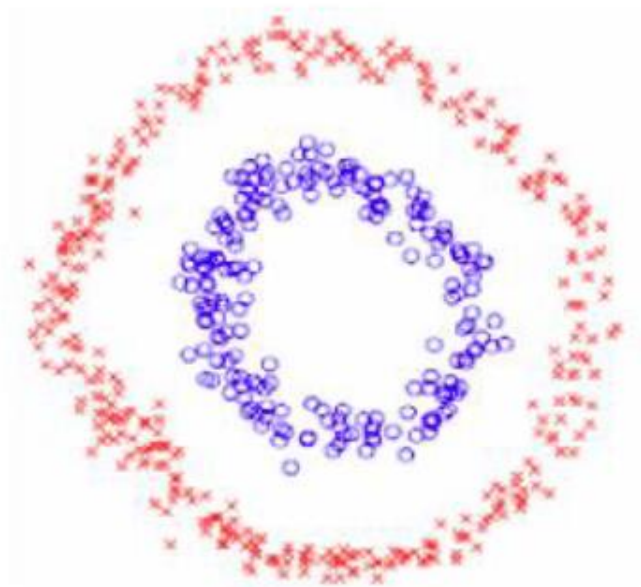
Several slides are credited to Prof. Aarti Singh, ML 10 – 701, CMU

Data Clustering

- Two different criteria
  - Compactness, e.g., k-means, mixture models
  - Connectivity, e.g., spectral clustering



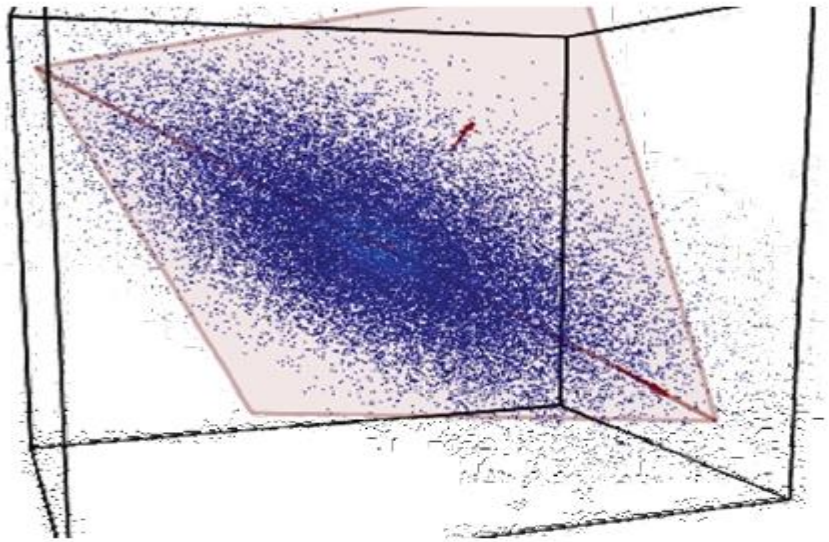
Compactness

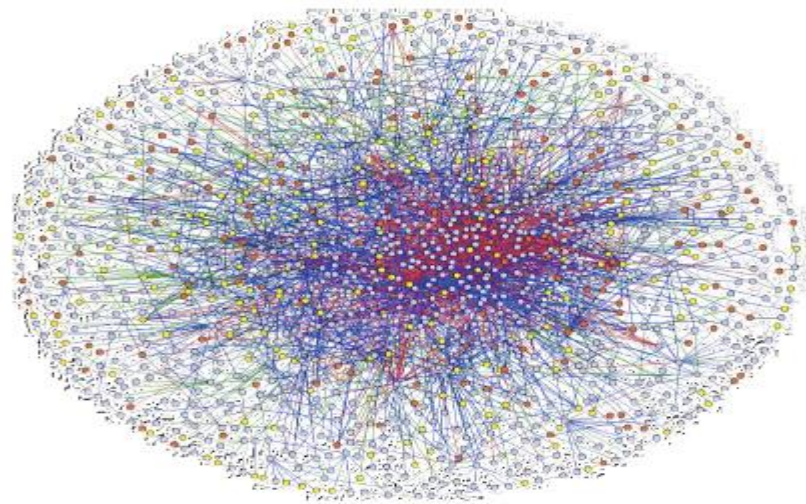


Connectivity

Connect Points in R^d and Graph Views of Data

- Points in R^d
  - via near-neighbor graphs
- Graph
  - via matrix representations of graphs



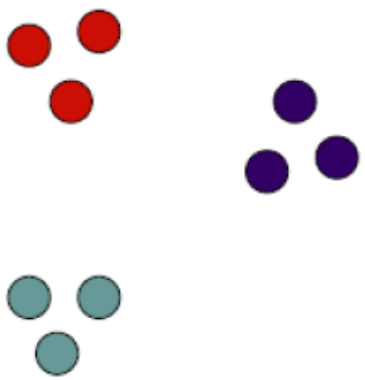


Graph Clustering

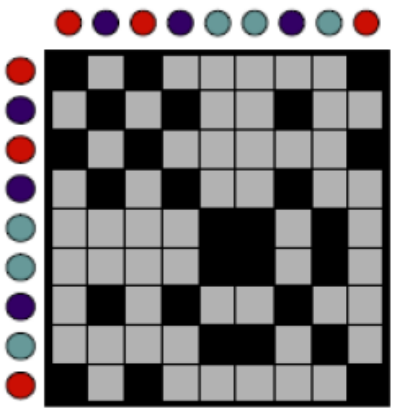
Goal: Given data points $X_1, \ldots, X_n$ and similarities $w(X_i,X_j)$, partition the data into groups so that points in a group are similar and points in different groups are dissimilar.

Similarity Graph: G(V,E,W)

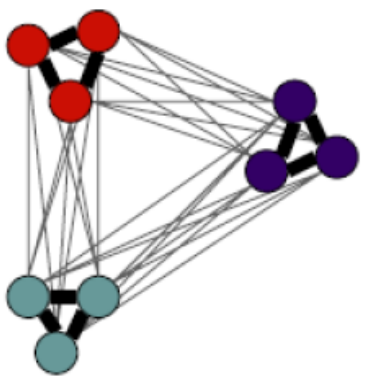
V – Vertices (Data points)
E – Edge if similarity > 0
W - Edge weights (similarities)



Data



Similarities

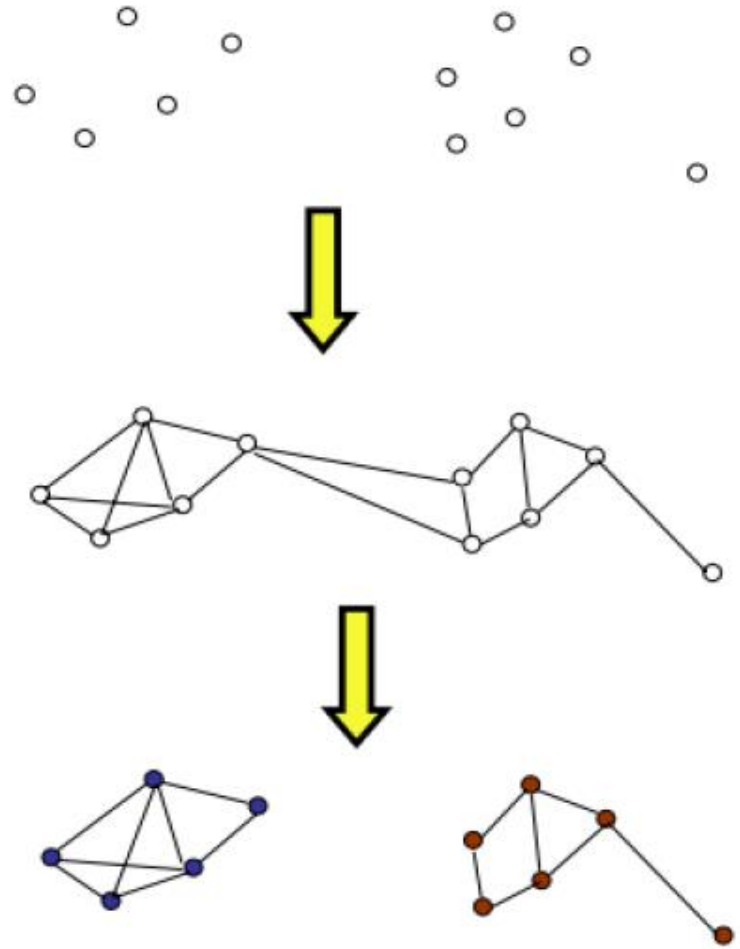


Similarity graph

Partition the graph so that edges within a group have large weights and edges across groups have small weights.

GENERAL

First - graph representation of data
(largely, application dependent)



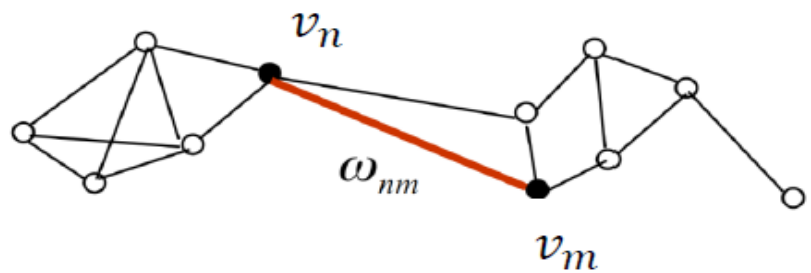
Then - graph partitioning

Disconnected graph components ⟷ Groups of points(Weakly connections in between components Strongly connections within components)

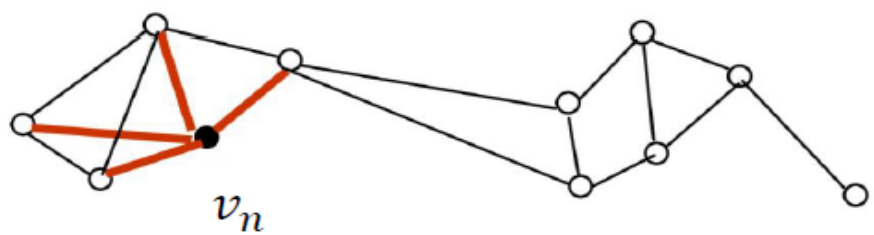
GRAPH NOTATION

$G=(V,E)$:

- Vertex set $V = \{v_1, \ldots, v_n\}$
- Weighted adjacency matrix $W = (w_{ij})\ i,j = 1, \ldots, n \quad w_{ij} \geq 0$



- Degree $d_i = \sum_{j=1}^{n} w_{ij}$



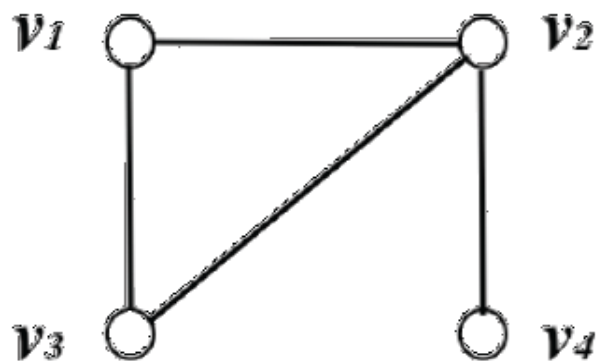
- Degree matrix Diagonal matrix with the degrees $d_1, \ldots, d_n$ on the diagonal.

Adjacency Matrices

- For a graph with n vertices, the entries of the $n \times n$ adjacency matrix are defined by:

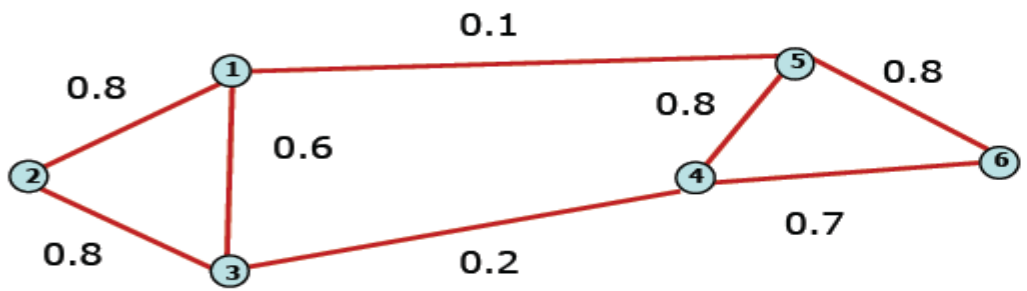
$$\mathbf{A} := \begin{cases} A_{ij} = 1 & \text{if there is an edge } e_{ij} \\ A_{ij} = 0 & \text{if there is no edge} \\ A_{ii} = 0 \end{cases}$$

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$



Weighted Matrices

- Adjacency matrix (A)
  - $n \times n$ matrix
  - $A = [w_{ij}]$: edge weight between vertex x_i and x_j



	x_1	x_2	x_3	x_4	x_5	x_6
x_1	0	0.8	0.6	0	0.1	0
x_2	0.8	0	0.8	0	0	0
x_3	0.6	0.8	0	0.2	0	0
x_4	0	0	0.2	0	0.8	0.7
x_5	0.1	0	0	0.8	0	0.8
x_6	0	0	0	0.7	0.8	0

- Important properties:
  - Symmetric matrix

  $\Rightarrow$ Eigenvalues are <u>real</u>

  $\Rightarrow$ Eigenvector could span <u>orthogonal base</u>

Eigenvalues and Eigenvectors

- $\mathbf{A}$ is a real-symmetric matrix: it has n real eigenvalues and its n real eigenvectors form an orthonormal basis.
- Let $\{\lambda_1, \ldots, \lambda_i, \ldots, \lambda_r\}$ be the set of *distinct* eigenvalues.
- The eigenspace S_i contains the eigenvectors associated with λ_i:

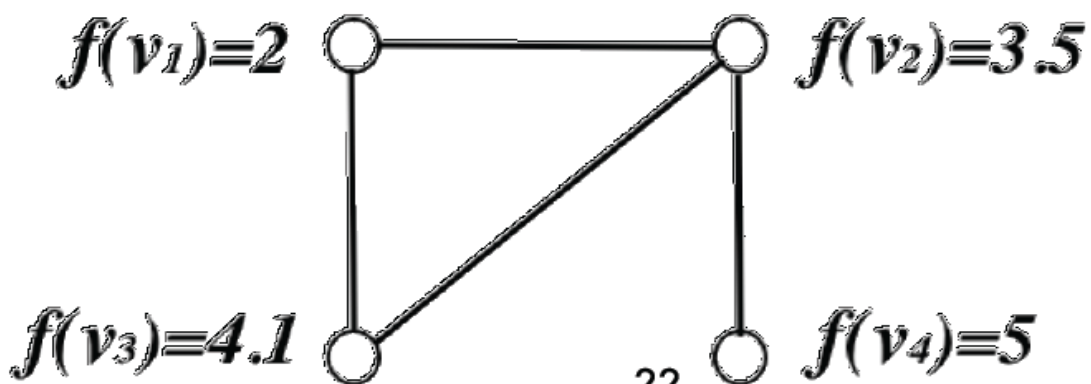
$$S_i = \{\boldsymbol{x} \in \mathbb{R}^n | \mathbf{A}\boldsymbol{x} = \lambda_i \boldsymbol{x}\}$$

- For real-symmetric matrices, the algebraic multiplicity is equal to the geometric multiplicity, for all the eigenvalues.
- The dimension of S_i (geometric multiplicity) is equal to the multiplicity of λ_i.
- If $\lambda_i \neq \lambda_j$ then S_i and S_j are mutually orthogonal.

Order the eigenvalues from small to large

Functions on Graphs

- We consider real-valued functions on the set of the graph's vertices, $\boldsymbol{f} : \mathcal{V} \longrightarrow \mathbb{R}$. Such a function assigns a real number to each graph node.
- $\boldsymbol{f}$ is a vector indexed by the graph's vertices, hence $\boldsymbol{f} \in \mathbb{R}^n$.
- Notation: $\boldsymbol{f} = (f(v_1), \ldots, f(v_n)) = (f(1), \ldots, f(n))$.
- The eigenvectors of the adjacency matrix, $\mathbf{A}\boldsymbol{x} = \lambda\boldsymbol{x}$, can be viewed as *eigenfunctions*.



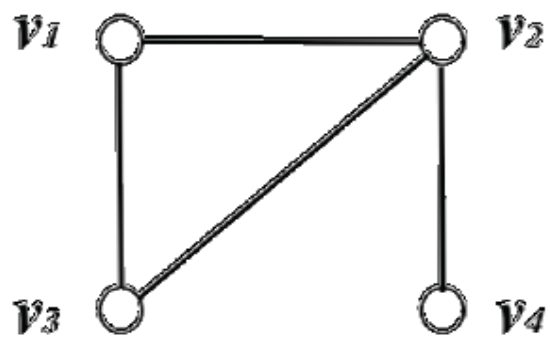
Graph (Unnormalized) Laplacian

- $\mathbf{L} = \nabla^\top \nabla$
- $(\mathbf{L}\boldsymbol{f})(v_i) = \sum_{v_j \sim v_i} (f(v_i) - f(v_j))$
- Connection between the Laplacian and the adjacency matrices:

$$\mathbf{L} = \mathbf{D} - \mathbf{A}$$

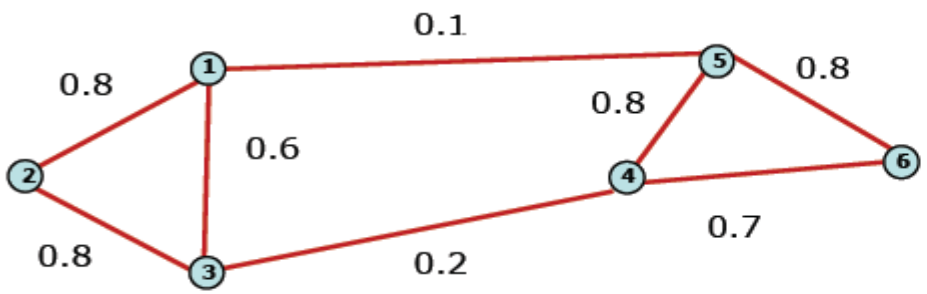
- The degree matrix: $\mathbf{D} := D_{ii} = d(v_i)$.

$$\mathbf{L} = \begin{bmatrix} 2 & -1 & -1 & 0 \\ -1 & 3 & -1 & -1 \\ -1 & -1 & 2 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}$$



Degree Matrix

- **Degree matrix (D)**
  - $n \times n$ diagonal matrix
  - $D(i,i) = \sum_j w_{ij}$: total weight of edges incident to vertex x_i



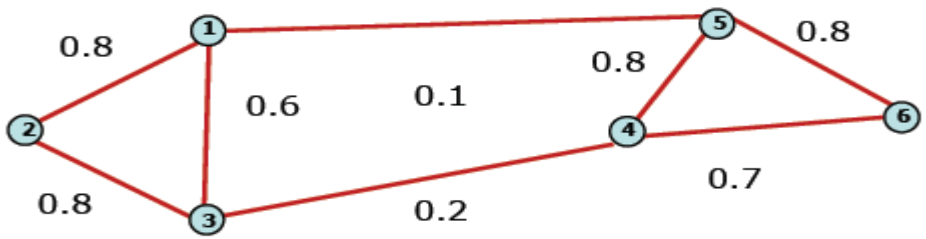
	x_1	x_2	x_3	x_4	x_5	x_6
x_1	1.5	0	0	0	0	0
x_2	0	1.6	0	0	0	0
x_3	0	0	1.6	0	0	0
x_4	0	0	0	1.7	0	0
x_5	0	0	0	0	1.7	0
x_6	0	0	0	0	0	1.5

- Important application:
  - Normalize adjacency matrix

Laplacian Matrix

- **Laplacian matrix (L)**
  - n x n symmetric matrix

$$L = D - A$$



	x_1	x_2	x_3	x_4	x_5	x_6
x_1	1.5	-0.8	-0.6	0	-0.1	0
x_2	-0.8	1.6	-0.8	0	0	0
x_3	-0.6	-0.8	1.6	-0.2	0	0
x_4	0	0	-0.2	1.7	-0.8	-0.7
x_5	-0.1	0	0	0.8-	1.7	-0.8
x_6	0	0	0	-0.7	-0.8	1.5

- Important properties:
  - Eigenvalues are non-negative real numbers (Gershgorin circle theorem)
  - Eigenvectors are real and orthogonal
  - Eigenvalues and eigenvectors provide an insight into the connectivity of the graph…

Undirected Weighted Graphs

- We consider *undirected weighted graphs*: Each edge e_{ij} is weighted by $w_{ij} > 0$.
- The Laplacian as an operator:

$$(\mathbf{L}\boldsymbol{f})(v_i) = \sum_{v_j \sim v_i} w_{ij}(f(v_i) - f(v_j))$$

- As a quadratic form:

$$\boldsymbol{f}^\top \mathbf{L} \boldsymbol{f} = \frac{1}{2} \sum_{e_{ij}} w_{ij}(f(v_i) - f(v_j))^2$$

- $\mathbf{L}$ is symmetric and positive semi-definite.
- $\mathbf{L}$ has n non-negative, real-valued eigenvalues: $0 = \lambda_1 \leq \lambda_2 \leq \ldots \leq \lambda_n$.

- Unnormalized Graph Laplacian

$$d_i = \sum_{j=1}^{n} w_{ij}$$

$$L = D - W$$

Proposition 1 (Properties of L) The matrix L satisfies the following properties:

1. For every $f \in \mathbb{R}^n$ we have

$$f'Lf = \frac{1}{2}\sum_{i,j=1}^{n} w_{ij}^2 (f_i - f_j)^2$$

$$f'Lf = f'Df - f'Wf = \sum_{i=1}^{n} d_i f_i^2 - \sum_{i,j=1}^{n} f_i f_j w_{ij}$$

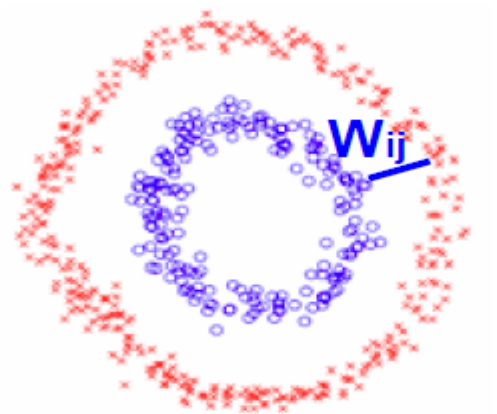
$$= \frac{1}{2}\left(\sum_{i=1}^{n} d_i f_i^2 - 2\sum_{i,j=1}^{n} f_i f_j w_{ij} + \sum_{j=1}^{n} d_j f_j^2\right) = \frac{1}{2}\sum_{i,j=1}^{n} w_{ij}(f_i - f_j)^2$$

Similarity graph construction

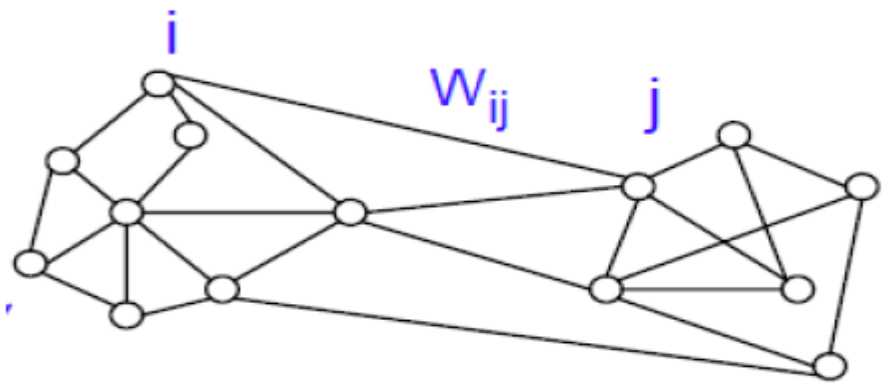
Similarity Graphs: Model local neighborhood relations between data points

E.g. Gaussian kernel similarity function

$$W_{ij} = e^{\frac{\|x_i - x_j\|^2}{2\sigma^2}}$$ → Controls size of neighborhood



Data clustering



$G = \{V,E\}$

SIMILARITY GRAPH

- ε-neighborhood graph

  Connect all points whose pairwise distances are smaller than ε

- k-nearest neighbor graph

  Connect vertex v_i with vertex v_j if v_j is among the k-nearest neighbors of v_i.

- fully connected graph

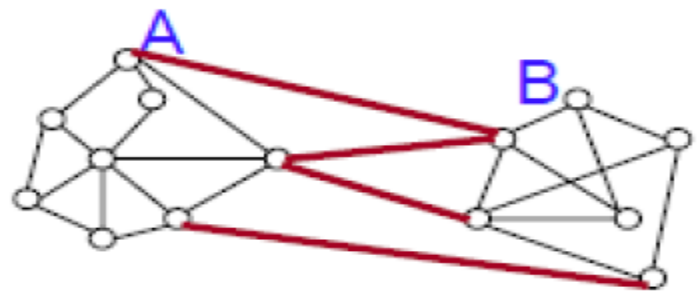
  Connect all points with positive similarity with each other

All the above graphs are regularly used in spectral clustering!

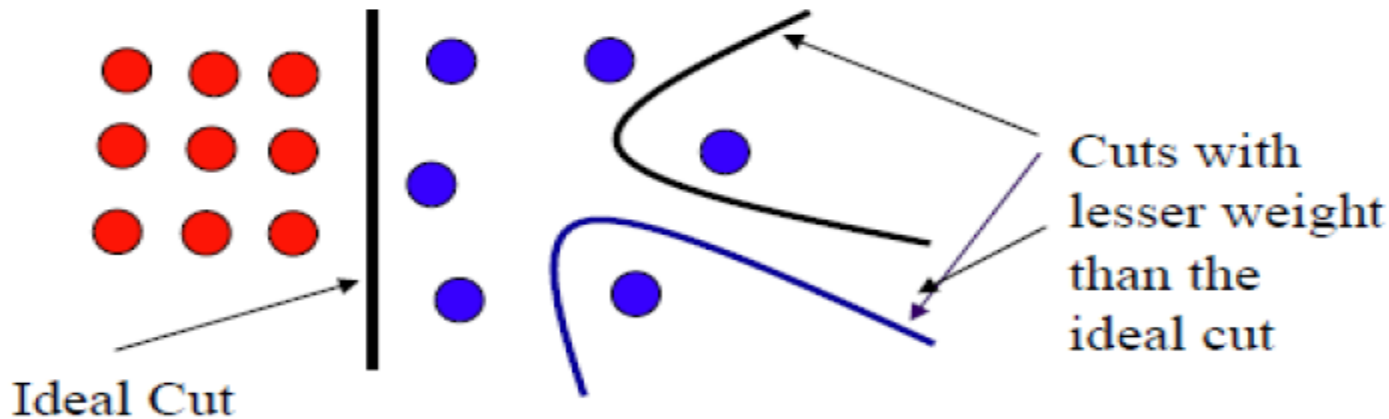
Partitioning a graph into two clusters

Min-cut: Partition graph into two sets A and B such that weight of edges connecting vertices in A to vertices in B is minimum.

$$\text{cut}(A, B) := \sum_{i \in A, j \in B} w_{ij}$$



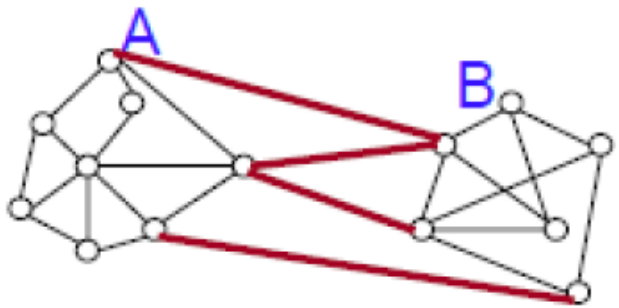
- Easy to solve O(VE) algorithm
- Not satisfactory partition – often isolates vertices



Partitioning a graph into two clusters

Partition graph into two sets A and B such that weight of edges connecting vertices in A to vertices in B is minimum & size of A and B are very similar.

$$\text{cut}(A, B) := \sum_{i \in A, j \in B} w_{ij}$$



Normalized cut:

$$\text{Ncut}(A, B) := \text{cut}(A, B)\left(\frac{1}{\text{vol}(A)} + \frac{1}{\text{vol}(B)}\right)$$

$$\text{vol}(A) = \sum_{i \in A} d_i$$

But NP-hard to solve!!

Spectral clustering is a relaxation of these.

Normalized Cut and Graph Laplacian

$$\text{Ncut}(A, B) := \text{cut}(A, B)\left(\frac{1}{\text{vol}(A)} + \frac{1}{\text{vol}(B)}\right)$$

Let $f = [f_1\ f_2\ \ldots\ f_n]^T$ with $f_i = \begin{cases} \frac{1}{\text{vol}(A)} & \text{if } i \in A \\ -\frac{1}{\text{vol}(B)} & \text{if } i \in B \end{cases}$

$$\mathbf{f}^T\mathbf{L}\mathbf{f} = \sum_{ij} w_{ij}(\mathbf{f}_i - \mathbf{f}_j)^2 = \sum_{i\in A, j\in B} w_{ij}\left(\frac{1}{\text{vol}(A)} + \frac{1}{\text{vol}(B)}\right)^2$$

$$\mathbf{f}^T\mathbf{D}\mathbf{f} = \sum_j d_i\mathbf{f}_i^2 = \sum_{i\in A}\frac{d_i}{\text{vol}(A)^2} + \sum_{j\in B}\frac{d_i}{\text{vol}(B)^2} = \frac{1}{\text{vol}(A)} + \frac{1}{\text{vol}(B)}$$

$$\text{Ncut}(A, B) = \frac{\mathbf{f}^T\mathbf{L}\mathbf{f}}{\mathbf{f}^T\mathbf{D}\mathbf{f}}$$

Normalized Cut and Graph Laplacian

$$\min \text{Ncut}(A, B) = \min \frac{\mathbf{f}^T \mathbf{L} \mathbf{f}}{\mathbf{f}^T \mathbf{D} \mathbf{f}}$$

where $\mathbf{f} = [f_1\ f_2\ \ldots\ f_n]^T$ with $f_i = \begin{cases} \frac{1}{\text{vol}(A)} & \text{if } i \in A \\ -\frac{1}{\text{vol}(B)} & \text{if } i \in B \end{cases}$

Relaxation: $\min \frac{\mathbf{f}^T \mathbf{L} \mathbf{f}}{\mathbf{f}^T \mathbf{D} \mathbf{f}}$ s.t. $\mathbf{f}^T \mathbf{D} \mathbf{1} = 0$

Solution: f – second eigenvector of generalized eval problem

$$\boxed{\mathbf{L}\mathbf{f} = \lambda \mathbf{D}\mathbf{f}}$$

Obtain cluster assignments by thresholding f at 0

Approximation of Normalized cut

$$\text{Ncut}(A, B) := \text{cut}(A, B)\left(\frac{1}{\text{vol}(A)} + \frac{1}{\text{vol}(B)}\right)$$

Let f be the eigenvector corresponding to the second smallest eval of the generalized eval problem.

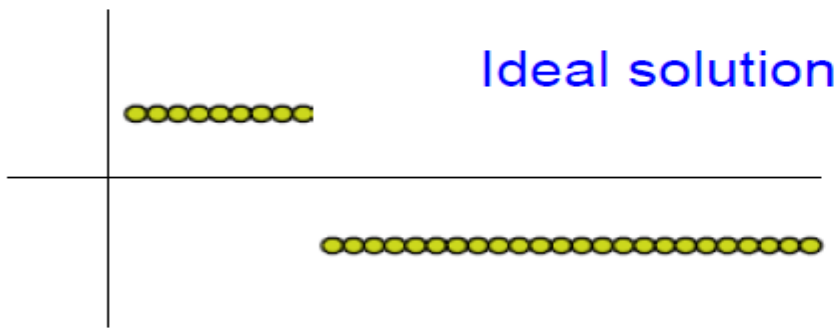
$$\mathbf{Lf} = \lambda \mathbf{Df}$$

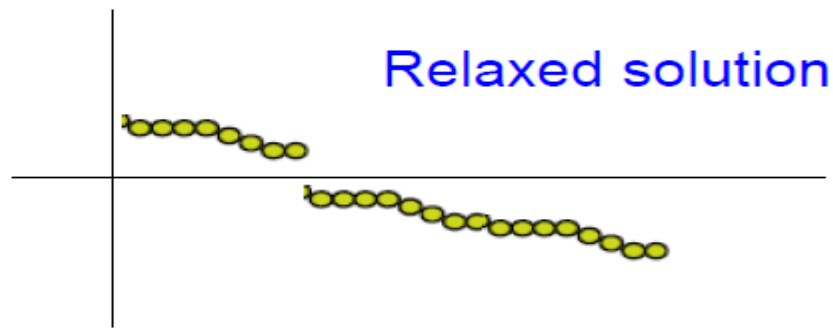
Equivalent to eigenvector corresponding to the second smallest eval of the normalized Laplacian $L' = D^{-1}L = I - D^{-1}W$

Recover binary partition as follows:

$i \in A$ *if* $f_i \geq 0$

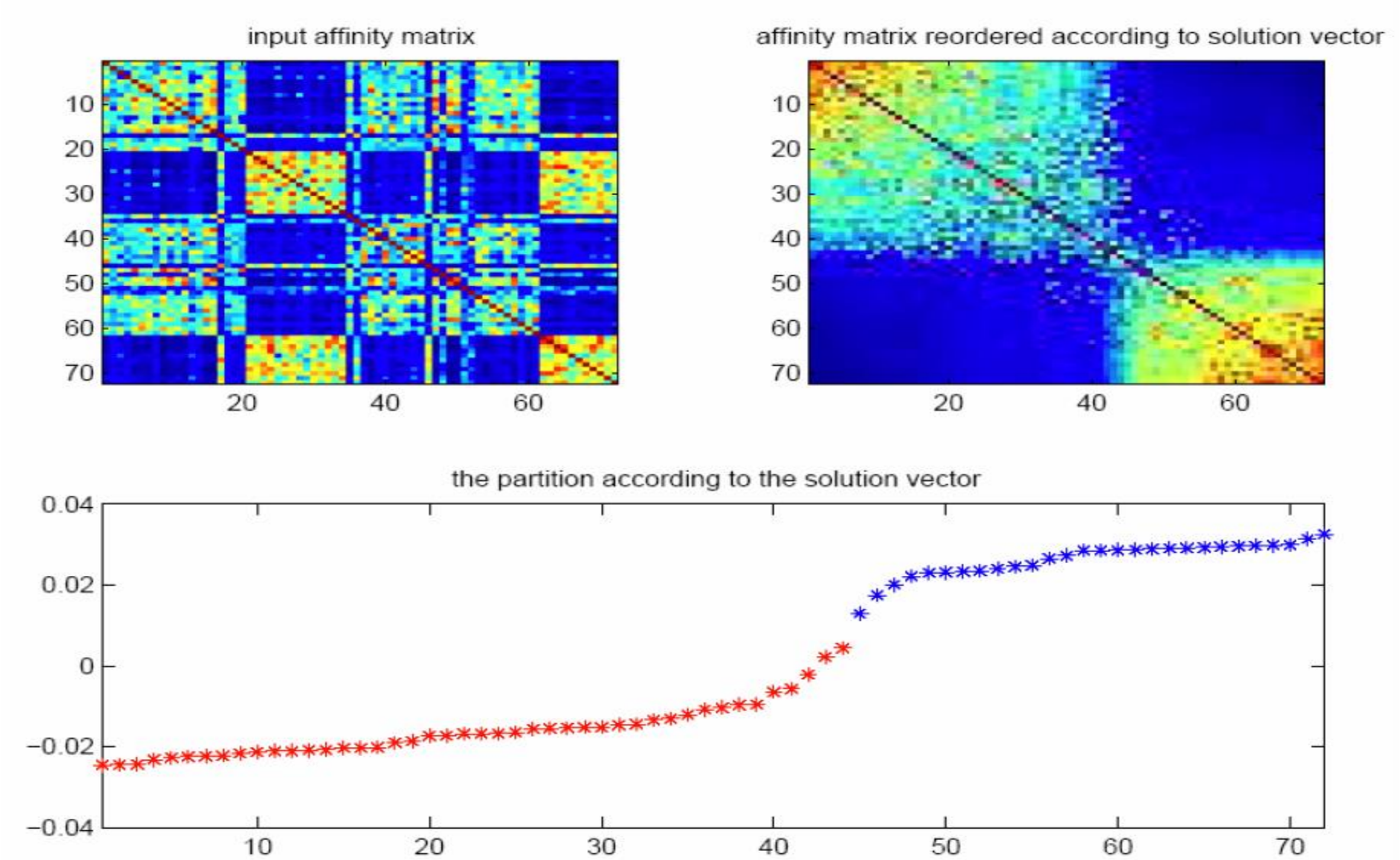
$i \in B$ *if* $f_i < 0$





Example

Xing et al 2001



How to partition a graph into k clusters?

Spectral Clustering Algorithm

Input: Similarity matrix W, number k of clusters to construct

- Build similarity graph
- Compute the first k eigenvectors $v_1, \ldots, v_k$ of the matrix

$$\begin{cases} L & \text{for unnormalized spectral clustering} \\ L' & \text{for normalized spectral clustering} \end{cases}$$

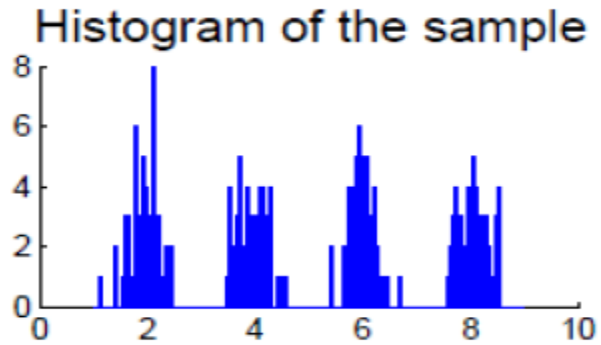
- Build the matrix $V \in \mathbb{R}^{n \times k}$ with the eigenvectors as columns
- Interpret the rows of V as new data points $Z_i \in \mathbb{R}^k$

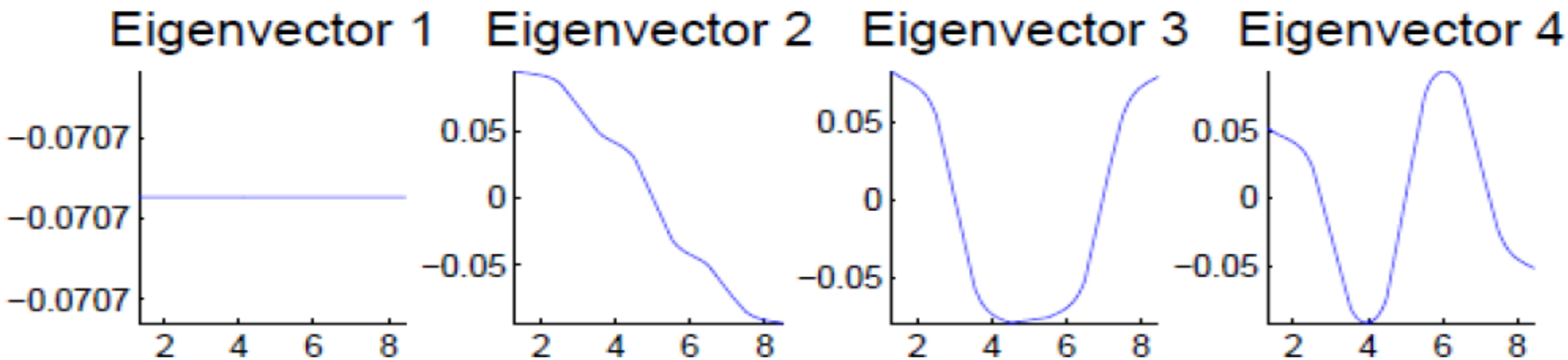
	v_1	v_2	v_3
Z_1	v_{11}	v_{12}	v_{13}
$\vdots$	$\vdots$	$\vdots$	$\vdots$
Z_n	v_{n1}	v_{n2}	v_{n3}

Dimensionality Reduction
n x n → n x k

- Cluster the points Z_i with the k-means algorithm in $\mathbb{R}^k$.

Eigenvectors of Graph Laplacian



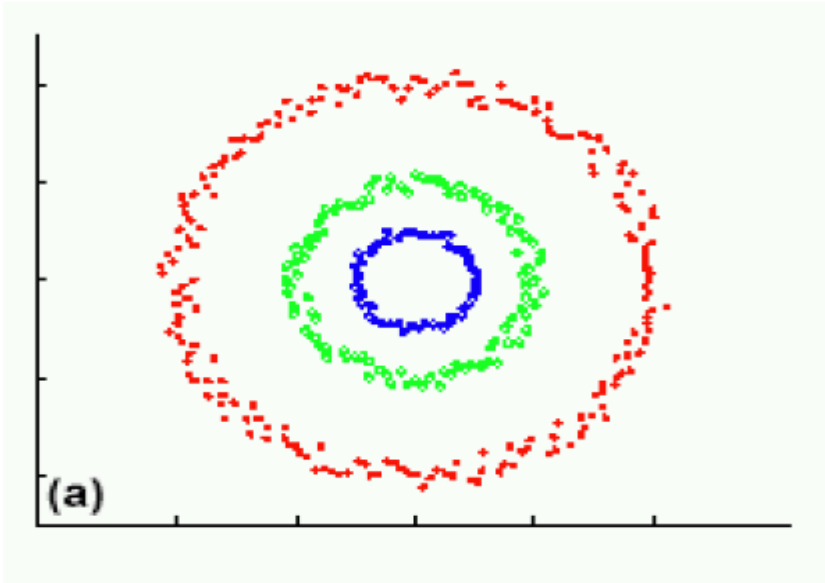


- 1st Eigenvector is the all ones vector ***1*** (if graph is connected)
- 2nd Eigenvector thresholded at 0 separates first two clusters from last two
- k-means clustering of the 4 eigenvectors identifies all clusters

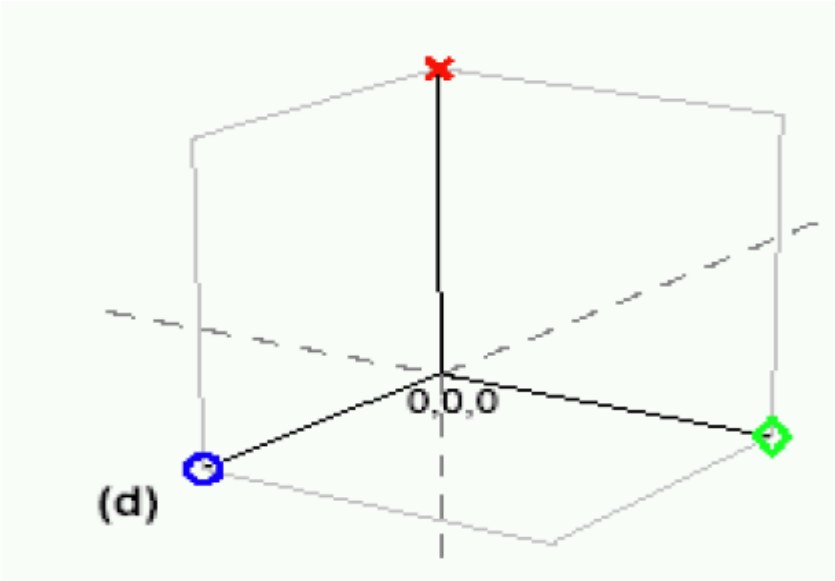
Why does it work?

Data are projected into a lower-dimensional space (the spectral/eigenvector domain) where they are easily separable, say using k-means.

Original data



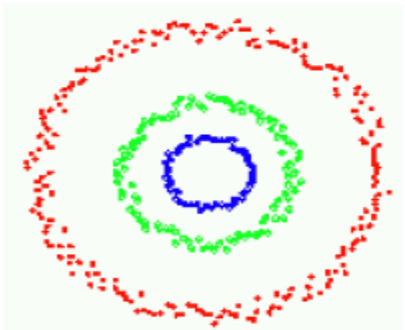
Projected data



Graph has 3 connected components – first three eigenvectors are constant (all ones) on each component.

Understanding Spectral Clustering

- If graph is connected, first Laplacian evec is constant (all 1s)
- If graph is disconnected (k connected components), Laplacian is block diagonal and first k Laplacian evecs are:



OR

$$L = \begin{bmatrix} L_1 & & 0 \\ & L_2 & \\ 0 & & L_3 \end{bmatrix}$$

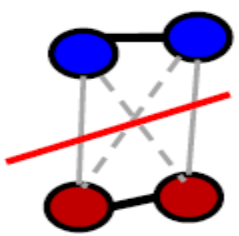
$$\begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

First three eigenvectors

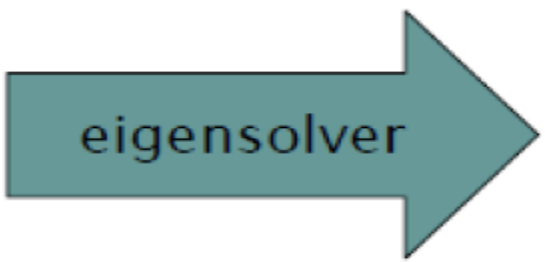
Understanding Spectral Clustering

- Is all hope lost if clusters don't correspond to connected components of graph? No!
- If clusters are connected loosely (small off-block diagonal enteries), then 1st Laplacian even is all 1s, but second evec gets first cut (min normalized cut)

$$\text{Ncut}(A, B) := \text{cut}(A, B)\left(\frac{1}{\text{vol}(A)} + \frac{1}{\text{vol}(B)}\right)$$



1	1	.2	0
1	1	0	.1
.2	0	1	1
0	.1	1	1



.50
.50
.50
.50

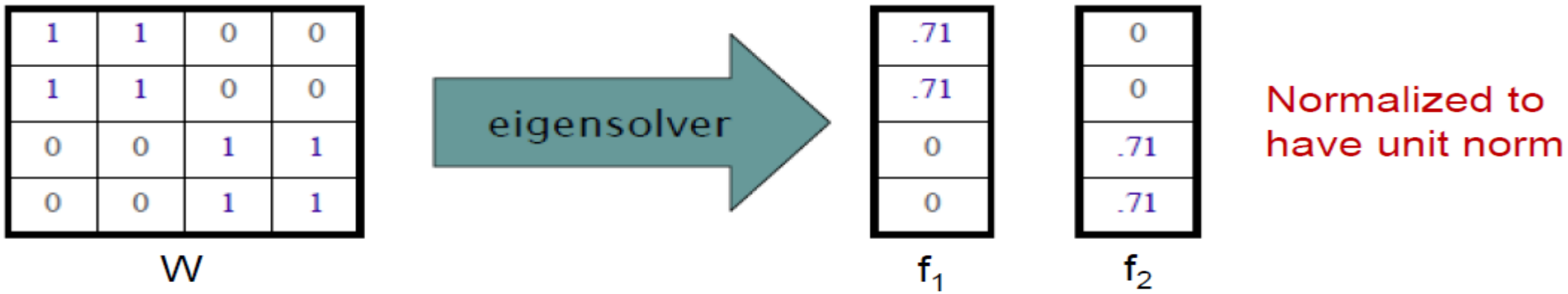
1st evec is constant since graph is connected

.47
.52
-.47
-.52

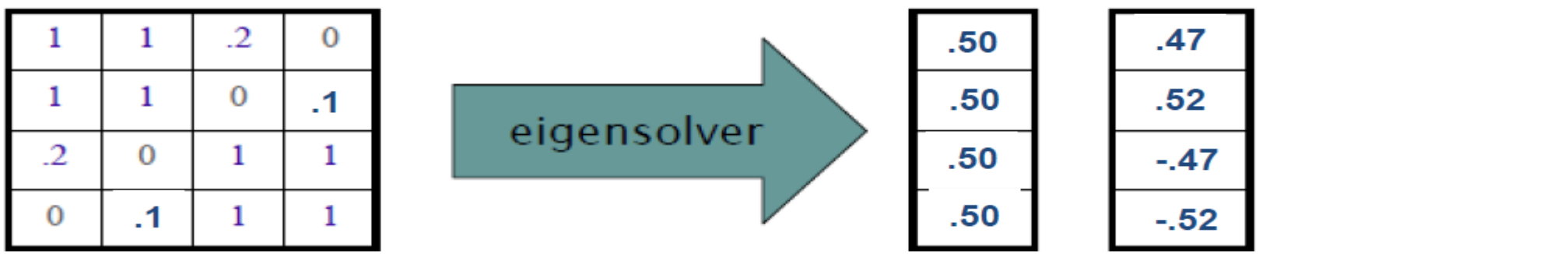
Sign of 2nd evec indicates blocks

Why does it work?

Block weight matrix (disconnected graph) results in block eigenvectors:



Slight perturbation does not change span of eigenvectors significantly:

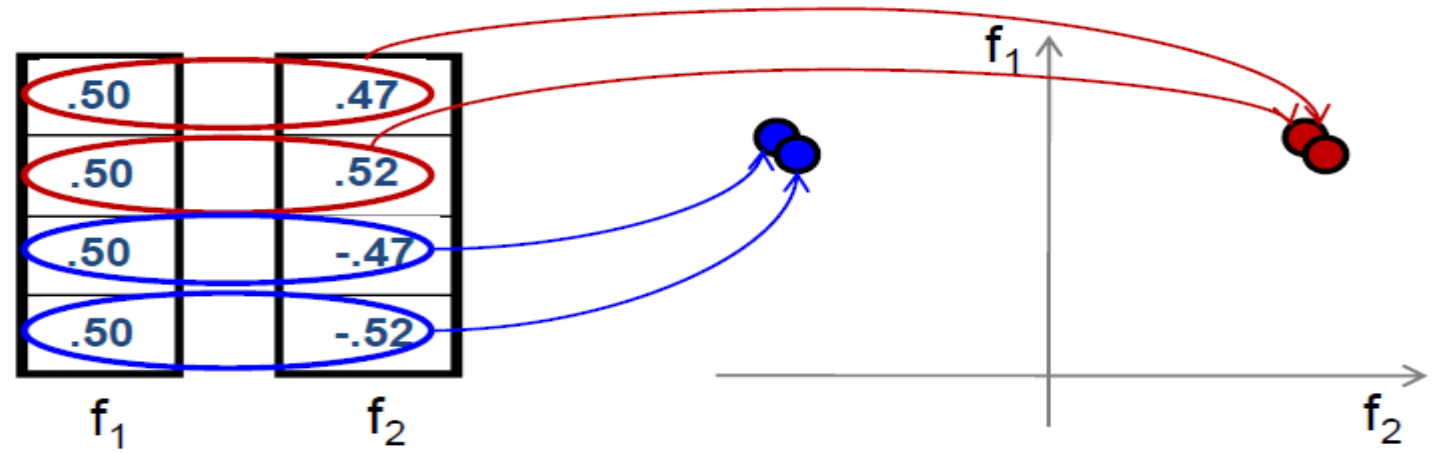


Why does it work?

Can put data points into blocks using eigenvectors:

1	1	.2	0
1	1	0	.1
.2	0	1	1
0	.1	1	1

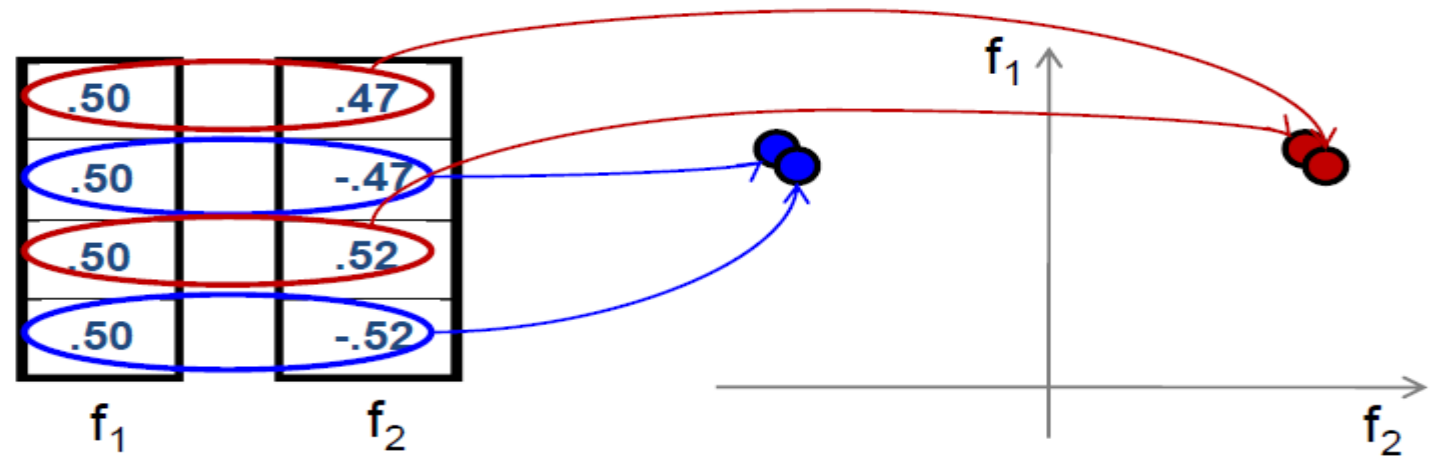
W



Embedding is same regardless of data ordering:

1	.2	1	0
.2	0	1	1
1	1	0	.1
0	1	.1	1

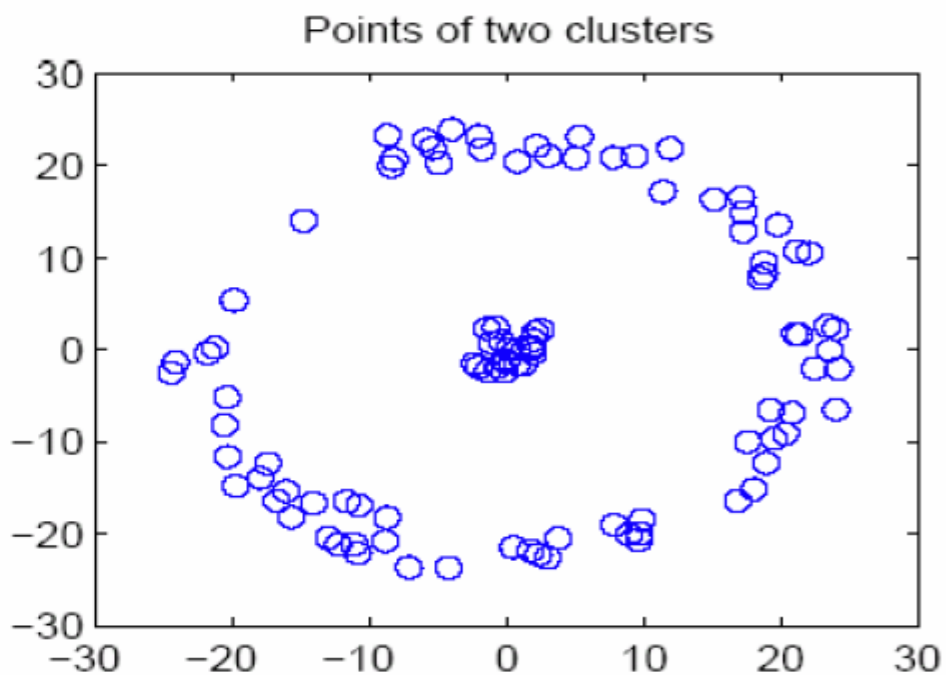
W



k-means vs Spectral clustering

Applying k-means to laplacian eigenvectors allows us to find cluster with non-convex boundaries.

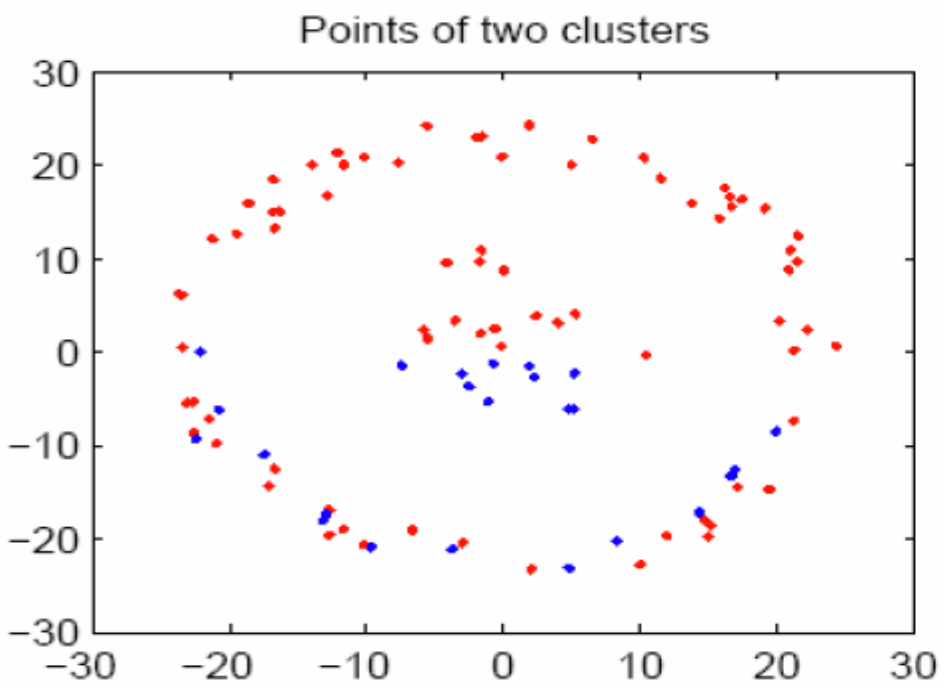
Both perform same



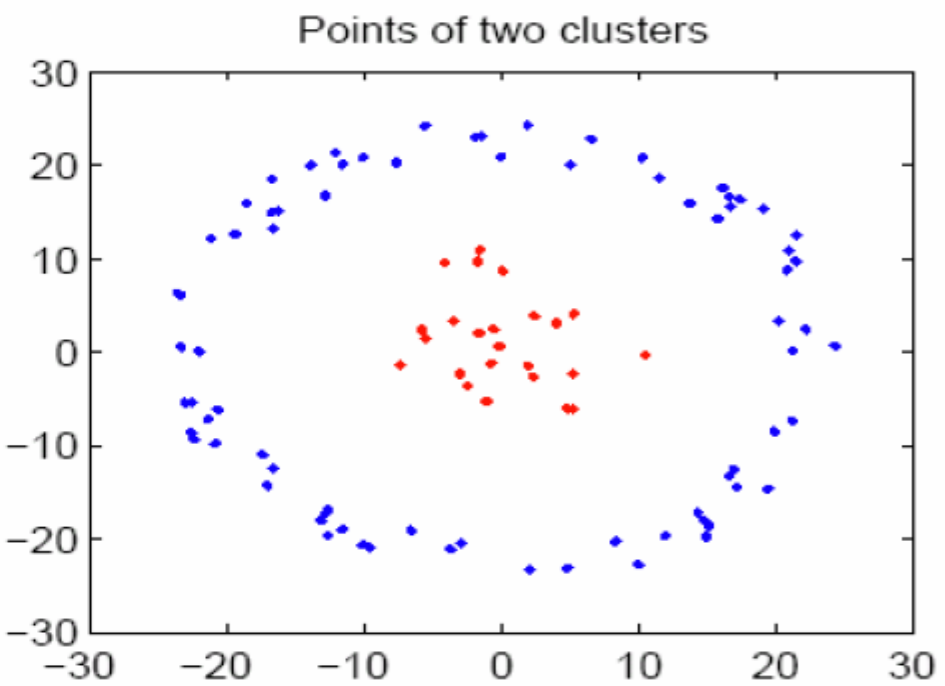
Spectral clustering is superior

k-means vs Spectral clustering

Applying k-means to laplacian eigenvectors allows us to find cluster with non-convex boundaries.



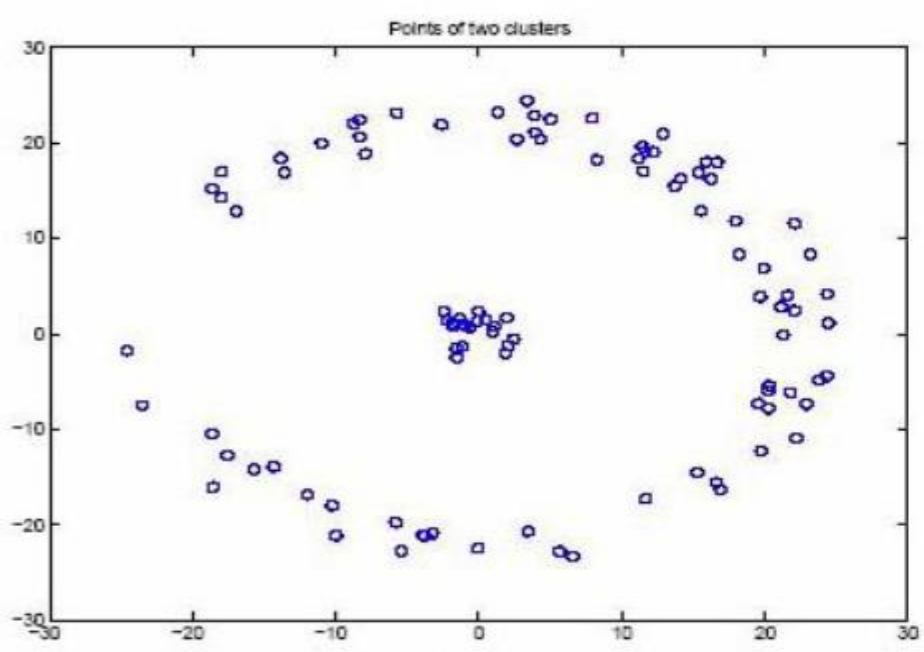
k-means output



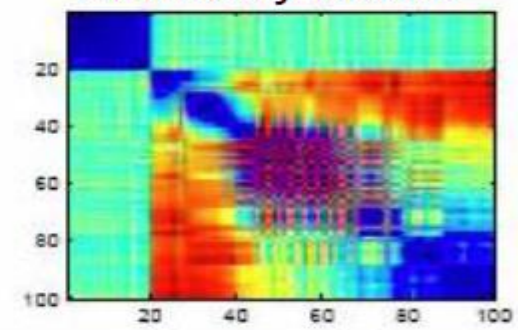
Spectral clustering output

k-means vs Spectral clustering

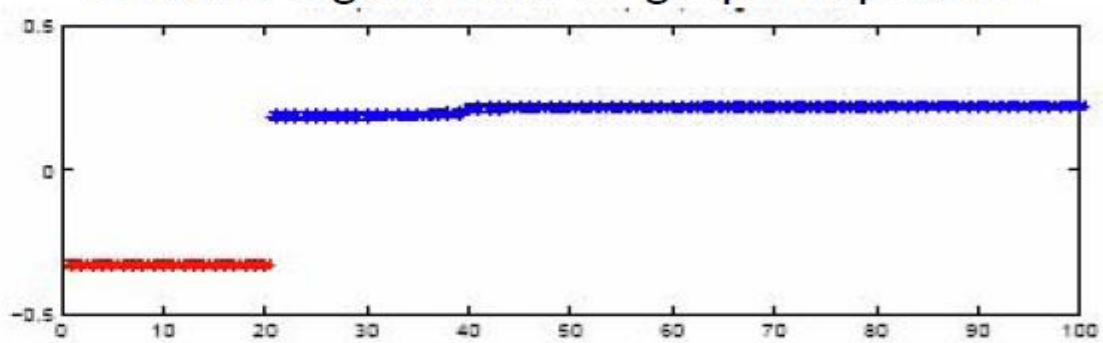
Applying k-means to laplacian eigenvectors allows us to find cluster with non-convex boundaries.



Similarity matrix

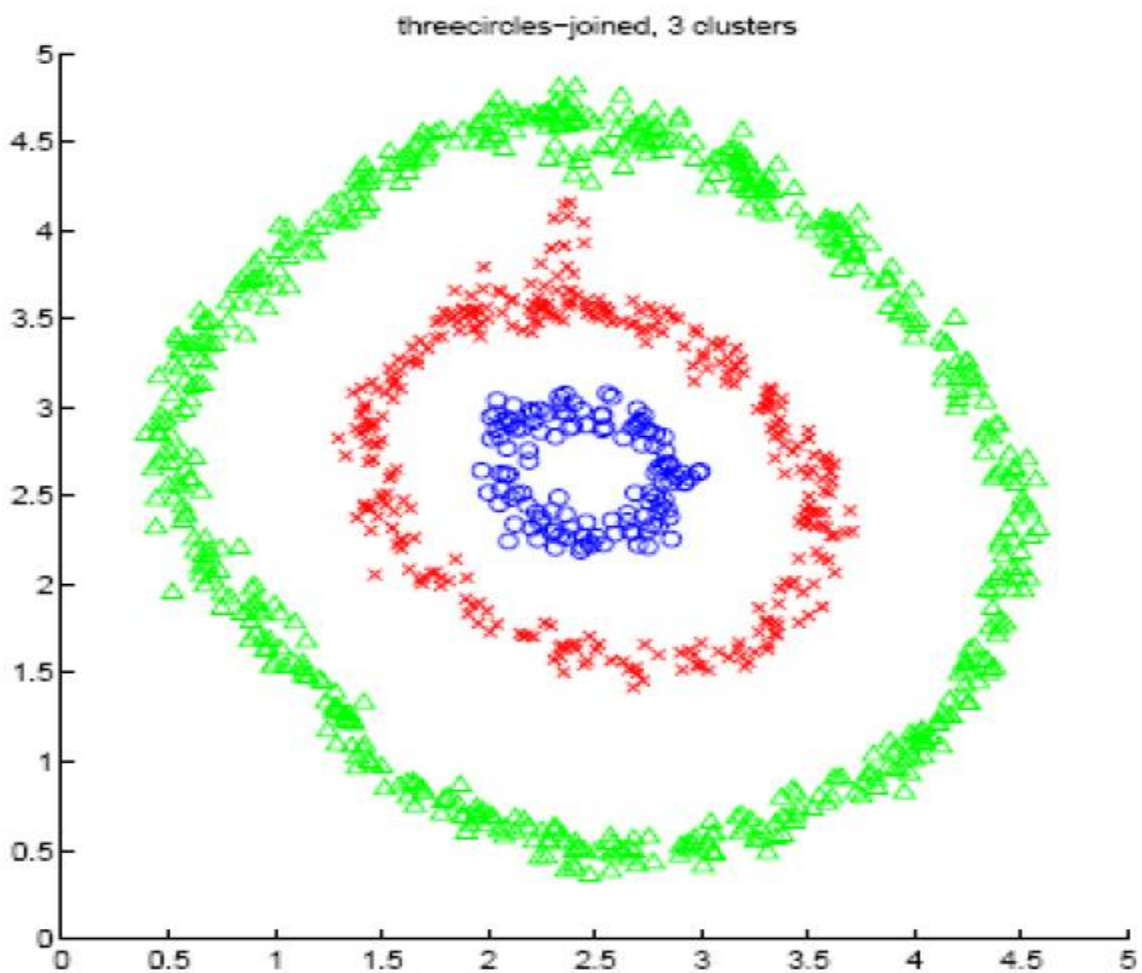


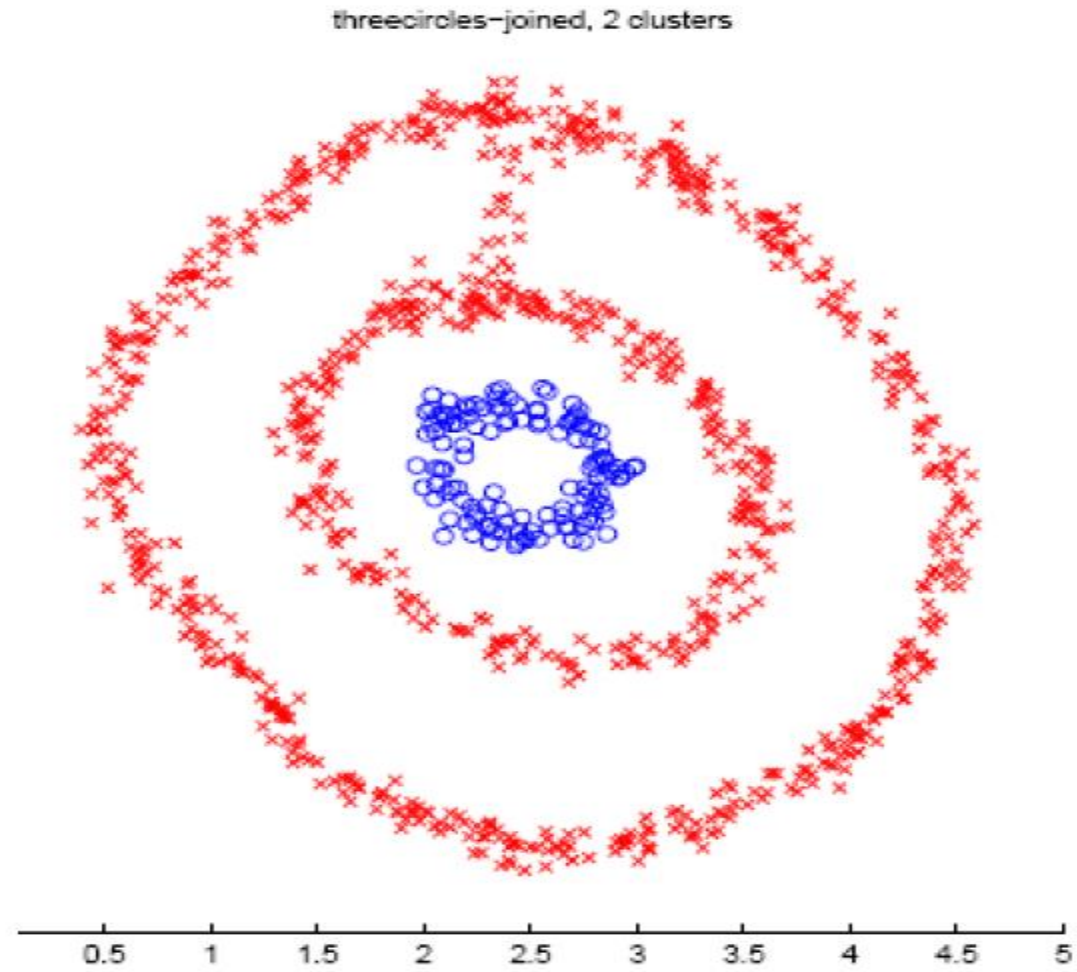
Second eigenvector of graph Laplacian



Examples (Choice of k)

Ng et al 2001





Spectral clustering summary

- Algorithms that cluster points using eigenvectors of matrices derived from the data
- Useful in hard non-convex clustering problems
- Obtain data representation in the low-dimensional space that can be easily clustered
- Variety of methods that use eigenvectors of unnormalized or normalized Laplacian, differ in how to derive clusters from eigenvectors, k-way vs repeated 2-way
- Empirically very successful

Connected Graph Laplacians

- $\mathbf{L}\boldsymbol{u} = \lambda \boldsymbol{u}$.
- $\mathbf{L}\mathbf{1}_n = \mathbf{0}$, $\lambda_1 = 0$ is the smallest eigenvalue.
- The *one* vector: $\mathbf{1}_n = (1 \dots 1)^\top$.
- $0 = \boldsymbol{u}^\top \mathbf{L} \boldsymbol{u} = \sum_{i,j=1}^{n} w_{ij}(u(i) - u(j))^2$.
- If any two vertices are connected by a path, then $\boldsymbol{u} = (u(1), \dots, u(n))$ needs to be constant at all vertices such that the quadratic form vanishes. Therefore, a graph with one connected component has the constant vector $\boldsymbol{u}_1 = \mathbf{1}_n$ as the only eigenvector with eigenvalue 0.

A Graph with k Connected Components

- Each connected component has an associated Laplacian. Therefore, we can write matrix $\mathbf{L}$ as a *block diagonal matrix*:

$$\mathbf{L} = \begin{bmatrix} \mathbf{L}_1 & & \\ & \ddots & \\ & & \mathbf{L}_k \end{bmatrix}$$

- The spectrum of $\mathbf{L}$ is given by the union of the spectra of $\mathbf{L}_i$.
- Each block corresponds to a connected component, hence each matrix $\mathbf{L}_i$ has an eigenvalue 0 with multiplicity 1.
- The spectrum of $\mathbf{L}$ is given by the union of the spectra of $\mathbf{L}_i$.
- The eigenvalue $\lambda_1 = 0$ has multiplicity k.

The Eigenspace of $\lambda_1 = 0$

- The eigenspace corresponding to $\lambda_1 = \ldots = \lambda_k = 0$ is spanned by the k mutually orthogonal vectors:

$$\begin{aligned} \boldsymbol{u}_1 &= \mathbf{1}_{L_1} \\ &\ldots \\ \boldsymbol{u}_k &= \mathbf{1}_{L_k} \end{aligned}$$

- with $\mathbf{1}_{L_i} = (000011111000)^\top \in \mathbb{R}^n$
- These vectors are the *indicator vectors* of the graph's connected components.
- Notice that $\mathbf{1}_{L_1} + \ldots + \mathbf{1}_{L_k} = \mathbf{1}_n$

The Fiedler Vector

- The first non-null eigenvalue λ_{k+1} is called the Fiedler value.
- The corresponding eigenvector $\boldsymbol{u}_{k+1}$ is called the Fiedler vector.
- The multiplicity of the Fiedler eigenvalue is always equal to 1.
- The Fiedler value is the *algebraic connectivity of a graph*, the further from 0, the more connected.
- The Fidler vector has been extensively used for *spectral bi-partioning*
- Theoretical results are summarized in Spielman & Teng 2007: `http://cs-www.cs.yale.edu/homes/spielman/`